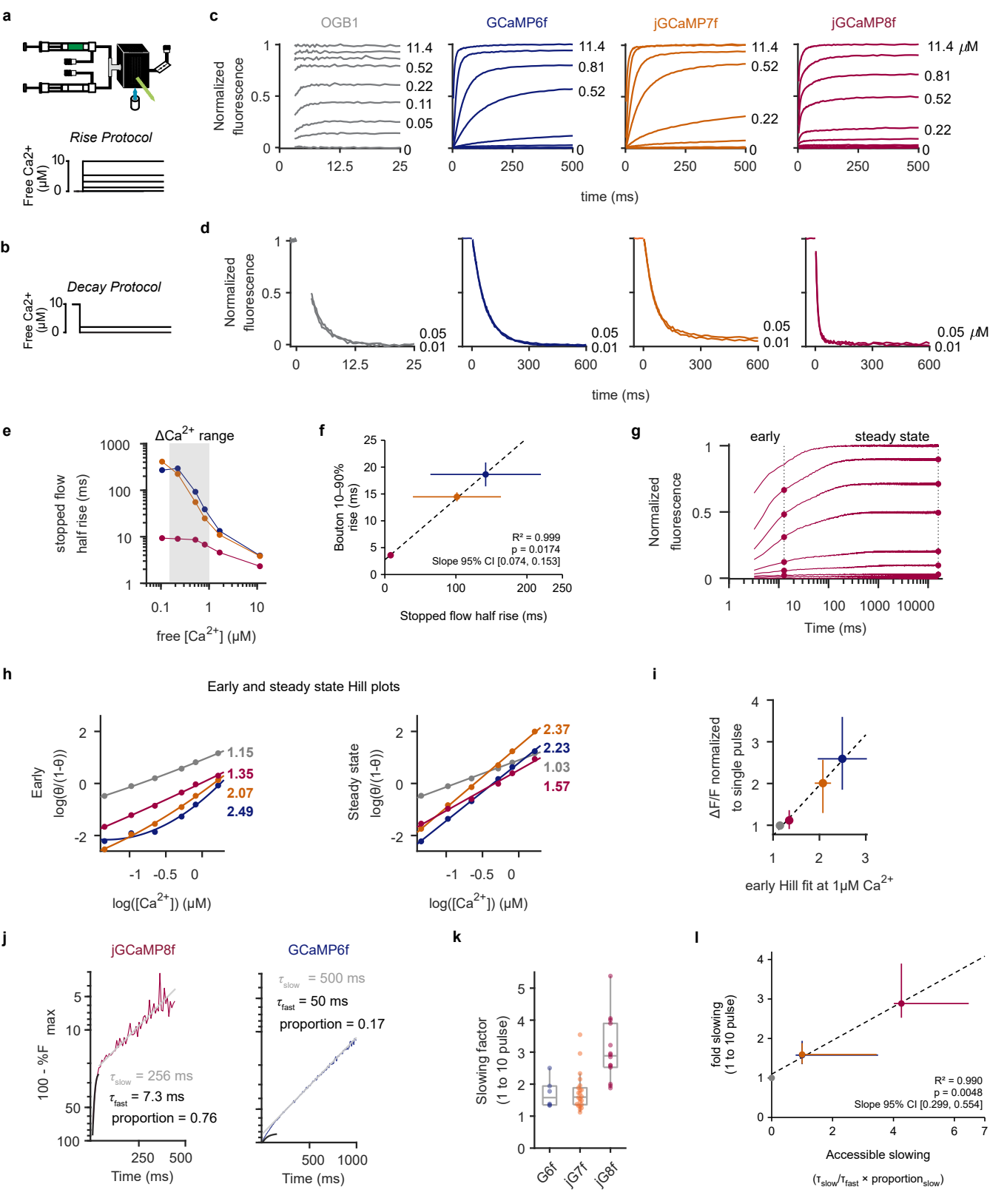

a
Rise Protocol
Free Ca2+ (μM)
10
0
b
Decay Protocol
Free Ca2+ (μM)
10
0
c
OGB1
GCaMP6f
jGCaMP7f
jGCaMP8f
Normalized fluorescence
11.4
0.52
0.22
0.11
0.05
0
11.4
0.81
0.52
0
11.4
0.52
0.22
0
11.4 μM
0.81
0.52
0.22
0
0
12.5
25
0
250
500
time (ms)
d
Normalized fluorescence
0.05
0.01
0
12.5
25
0
300
600
time (ms)
e
ΔCa2+ range
stopped flow half rise (ms)
1000
100
10
1
0.1
1
10
free [Ca2+] (μM)
f
Bouton 10–90% rise (ms)
25
20
15
10
5
0
R² = 0.999
p = 0.0174
Slope 95% CI [0.074, 0.153]
0
100
200
250
Stopped flow half rise (ms)
g
early
steady state
Normalized fluorescence
1
0.5
0
1
10
100
1000
10000
Time (ms)
h
Early and steady state Hill plots
Early
log(θ/(1-θ))
2
0
-2
1.15
1.35
2.07
2.49
-1
-0.5
0
log([Ca2+]) (μM)
Steady state
log(θ/(1-θ))
2
0
-2
2.37
2.23
1.03
1.57
-1
-0.5
0
log([Ca2+]) (μM)
i
ΔF/F normalized to single pulse
4
3
2
1
1
2
3
early Hill fit at 1μM Ca2+
j
jGCaMP8f
100 - %F max
5
10
50
100
τslow = 256 ms
τfast = 7.3 ms
proportion = 0.76
250
500
Time (ms)
GCaMP6f
τslow = 500 ms
τfast = 50 ms
proportion = 0.17
500
1000
Time (ms)
k
Slowing factor (1 to 10 pulse)
5
4
3
2
1
0
G6f
jG7f
jG8f
l
fold slowing (1 to 10 pulse)
4
3
2
1
0
R² = 0.990
p = 0.0048
Slope 95% CI [0.299, 0.554]
0
2
4
6
7
Accessible slowing
(τslow/τfast × proportionslow)